

PUPIL NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 5 • 1914–1918

The British Sector of the Western Front, 1914–1918

Injuries, Treatment and the Trenches in the British Sector



Primary Visual Source: RAMC Stretcher Bearers at Passchendaele • Flanders Clay (1917)

Accession Shelfmark: IWM-Q-2757

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE	LEARNT	REVISED
Key Topic 5.1: The Theatre of War: The British Sector, Trench Geography & Battles <i>Why were trenches designed in zig-zag traverses, and how did terrain at Ypres, Somme, and Arras affect medical treatment?</i>	☐	☐
Key Topic 5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses <i>How did frontline conditions cause trench foot, trench fever, and dysentery, and what preventive routines were enforced?</i>	☐	☐
Key Topic 5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection <i>Why did explosive artillery produce gas gangrene, and how did the Brodie steel helmet and respirators reduce fatalities?</i>	☐	☐
Key Topic 5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS <i>How did the RAMC and FANY evacuate wounded soldiers through six stations, and why was triage critical at CCSs?</i>	☐	☐
Key Topic 5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays <i>How did the Thomas Splint slash femur mortality from 80% to 20%, and why was Carrel-Dakin irrigation necessary?</i>	☐	☐
Key Topic 5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction <i>How did Oswald Robertson establish the first blood bank at Cambrai, and how did Gillies rebuild shattered faces at Sidcup?</i>	☐	☐

LIVING TIMELINE • PART 1: FRONTLINE CONDITIONS & EARLY TRAUMA (1914–1915)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

OCT–NOV 1914 • 1st Battle of Ypres & The Onset of Static Trench Warfare

Key Topic 5.1

The mobile war ends as both armies dig continuous defensive lines from the English Channel to Switzerland. British troops defend the vulnerable, low-lying Ypres Salient, surrounded on three sides by German artillery on high ground, with an impenetrable clay water table just two feet below the surface.

1915 • Chlorine Gas at 2nd Ypres & The Brodie Steel Helmet

Key Topic 5.2 • 5.3

In April 1915, the German Army launches the first chemical chlorine gas cloud attack at 2nd Ypres, causing fatal pulmonary oedema. Soldiers improvise urine pads before British PH flannel hoods are issued. John Brodie patents the pressed-steel helmet in late 1915, slashing fatal head wounds by 80%.

DEC 1915 • Robert Jones Introduces the Thomas Splint

Key Topic 5.5

Compound femur fractures from artillery shrapnel carry an 80% death rate due to muscle spasm driving bone ends through femoral arteries. Robert Jones trains frontline RAMC officers to fit Hugh Owen Thomas’s splint at the aid post, applying mechanical traction to pull bone ends straight and slashing mortality to 20%.

LIVING TIMELINE • PART 2: SURGICAL INNOVATION & BLOOD STORAGE (1916–1918)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

JULY 1916 • Battle of the Somme, Debridement & Carrel-Dakin Irrigation

Key Topic 5.1 • 5.5

The British Army suffers 57,470 casualties on Day 1 of the Somme, overwhelming Casualty Clearing Stations. Surgeons abandon Listerian superficial antiseptics for wide wound debridement (excision) to excise dead necrotic tissue before anaerobic gas gangrene develops, coupled with continuous Carrel-Dakin bleach irrigation.

1917 • Arras Underground Hospital & The Mud Logistics of Passchendaele

Key Topic 5.1 • 5.4

At Arras, New Zealand tunnelers excavate Thompson's Cave—a 700-bed underground hospital with operating theatres, running water, and electric lights 20m beneath the chalk. At 3rd Ypres (Passchendaele), artillery destroys all drainage systems, creating knee-deep liquid mud requiring 6 bearers per stretcher.

NOV 1917 • Oswald Robertson's Blood Depot & Gillies Facial Reconstruction

Key Topic 5.6

At the Battle of Cambrai, Captain Oswald Robertson deploys the world's first blood bank, using sodium citrate to stop clotting, glucose to preserve red cells, and ice chests to store Group O blood for 28 days. In Britain, Harold Gillies pioneers plastic facial surgery using tubed pedicles at Queen's Hospital, Sidcup.

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In which century BC did Hippocrates practise clinical observation?	2. Name the four bodily fluids in humoral medicine.
.....	
3. Which Roman physician developed the Theory of Opposites?	4. In what year did the Black Death reach England?
.....	
5. Which Renaissance anatomist published De Fabrica in 1543?	6. Who proved blood circulates through the body in 1628?
.....	
7. Which physician classified diseases through bedside observation?	8. In what year was the Royal Society founded?
.....	
9. What term describes foul air believed to cause disease?	10. What religious institution monopolized medical copying in the Middle Ages?
.....	

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Write a concise historical explanation demonstrating the tactical medical difference between a communication trench and a sap.

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.....

.....

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

1(a) Describe one feature of the system of communication trenches on the Western Front. [2 marks]

Hint: Consider how communication trenches connected frontline trenches to dressing stations and supply lines.

Feature:	Detail:
.....	

1(b) Describe one feature of the underground hospital at Arras (Thompson's Cave). [2 marks]

Hint: Consider the chalk geology, lighting, water supply, and surgical capacity 20 metres underground.

Feature:	Detail:
.....	

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY ASSESSMENT

SOURCE A: AERIAL RECONNAISSANCE PHOTOGRAPH OF BRITISH TRENCH SYSTEM NEAR YPRES (1917)

SOURCE-A-AERIAL-TRENCH-YPRES



Royal Flying Corps aerial photograph showing the intricate zig-zag traverses of frontline, support, and communication trenches near Ypres, criss-crossed with artillery shell craters.

Provenance: Official military aerial photograph taken by the Royal Flying Corps, British Sector near Ypres, 1917. Imperial War Museum Collection (Shelfmark: Q 42231).

Historical Context: An authentic aerial reconnaissance photograph taken by the Royal Flying Corps in 1917 showing the complex defensive network of British frontline, support, and communication trenches in the Ypres sector, displaying the characteristic zig-zag traverse design and dense shell cratering.

Hinge Question: Study Source A. How does the zig-zag design of the trenches shown in the photograph explain how British defensive engineers sought to protect soldiers from artillery shrapnel and machine-gun fire?

Study Source A. How useful is Source A for an enquiry into the defensive layout and tactical design of trench systems on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an official military aerial reconnaissance photograph taken by the Royal Flying Corps in 1917. What does an overhead view reveal about traverses, fire-steps, and saps, and what ground-level medical and mud conditions does it omit?

STRUCTURE STRIP • 3-STEP UTILITY ANALYSIS

1. CONTENT & UTILITY

Explain how the aerial photograph clearly shows the zig-zag traverse design, communication lines, and support saps.

2. PROVENANCE & MOTIVE

Evaluate the RFC photographic intelligence: highly accurate for cartography, but taken from high altitude under combat conditions.

3. CONTEXT & LIMITATIONS

CROSS-REFERENCE: Cross-reference with knowledge of mud, water tables at Ypres, and the difficulty stretcher bearers faced turning sharp right-angle corners.

ANALYTICAL CONNECTIVES:

Source A is useful for an enquiry into trench layout because... • Specifically, the aerial perspective reveals... • In terms of provenance, as an official RFC reconnaissance photograph... • However, the source is limited because it cannot show... • Therefore...

KEY VOCABULARY BANK:

Royal Flying Corps • zig-zag traverses • communication trench • Ypres Salient
• Messines Ridge • water table • duckboards • parapet • parados • fire-step

Task: Using the structure strip above, write your analytical source utility evaluation below:

TIMELINE MISSION

Sketch and annotate the zig-zag traverse pattern and low-lying Ypres water table on Milestone 1

← Pages 2-3

Trench drainage tip: If your duckboards are floating, you are no longer in an infantry trench; you are commanding a submarine.

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. Which French chemist published the Germ Theory of Disease in 1861?	2. Which German scientist identified the specific bacteria for anthrax (1876) and TB (1882)?
3. Who discovered the smallpox vaccination in 1796?	4. Which pioneer transformed sanitation at Scutari Hospital in 1854?
5. Which Scottish doctor discovered the anaesthetic properties of chloroform in 1847?	6. Who pioneered antiseptic carbolic acid spray in surgery in 1865?
7. Who proved cholera was waterborne via the Broad Street pump in 1854?	8. What year was the Second Public Health Act passed in Britain?
9. What was the primary danger of static warfare in the low-lying Ypres Salient?	10. Why were trenches dug in zig-zag traverses rather than straight lines?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain how trench foot differed biologically from trench fever, identifying the distinct cause and preventive measure for each.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

1(a) Describe one feature of the methods used to prevent trench foot on the Western Front. [2 marks]

Hint: Consider the buddy-system inspections, grease rubbing routines, and spare sock requirements.

Feature:	Detail:
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1(b) Describe one feature of the causes of trench fever among British troops. [2 marks]

Hint: Identify the parasite vector (body lice) and how lice faeces entered scratched skin.

Feature:	Detail:
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Question 2(b): Follow-Up Enquiry Table [4 marks • 5 PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY mins]

SOURCE A: SOLDIERS OF THE CHESHIRE REGIMENT GREASING FEET IN A RESERVE TRENCH (1916)

SOURCE-A-CHESHIRE-REGIMENT-TRENCH



Provenance: Official British War Office photograph taken on the Western Front, 1916. Imperial War Museum Collection (Shelfmark: Q 5156).

Historical Context: An authentic wartime photograph taken in 1916 showing British infantrymen of the Cheshire Regiment carrying out the mandatory daily foot inspection and greasing routine using whale oil to prevent the onset of trench foot in cold, damp conditions.

Hinge Question: Study Source A. How does this photograph demonstrate the organized preventive measures enforced by the British Army to stop soldiers from developing trench foot?

British infantrymen of the 8th Battalion, Cheshire Regiment, seated on the firestep of a trench, removing their boots and rubbing whale oil onto their feet under the supervision of a non-commissioned officer.

Study Source A. How could you follow up Source A to find out more about the methods used by the British Army to prevent trench foot on the Western Front?

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

SENTENCE STEMS:

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

KEY VOCABULARY BANK:

KEY VOCABULARY BANK:
trench foot • whale oil • buddy-system • gangrene • trench fever • body lice
(*Pediculus humanus*) • delousing stations • dysentery • chloride of lime •
latrine inspection

Enquiry Rationale: Explain in full sentences why your chosen source type is historically reliable and superior to alternatives:

TIMELINE MISSION

Draw soldiers applying whale oil and delousing garments on Milestone 2

← Pages 2-3

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. What percentage of Western Front wounds were caused by artillery shells and shrapnel?

3. In what year was the British pressed-steel Brodie helmet first issued?

5. In which battle did the German Army first deploy chlorine poison gas in April 1915?

7. Which gas introduced in 1917 caused internal and external blistering and blindness?

9. Who proved that cholera was caused by contaminated water rather than miasma?
2. Which bacterium in heavily manured Flanders soil caused fatal gas gangrene?

4. By what percentage did the Brodie helmet reduce fatal shrapnel head wounds?

6. What emergency measure did British troops use before gas masks were invented?

8. What antiseptic chemical did Joseph Lister pioneer in 1865?

10. What was the daily routine used in British trenches to prevent trench foot?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain the crucial medical difference between wound contamination from shrapnel and chemical tissue destruction from mustard gas.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

1(a) Describe one feature of the effects of poison gas attacks on soldiers on the Western Front. [2 marks]

Hint: Distinguish between chlorine (suffocation/pulmonary oedema), phosgene, and mustard gas (blistering).

Feature:

Detail:

1(b) Describe one feature of the design of the British Brodie steel helmet. [2 marks]

Hint: Consider the pressed-steel construction, shallow brim, and internal lining to deflect shrapnel.

Feature:

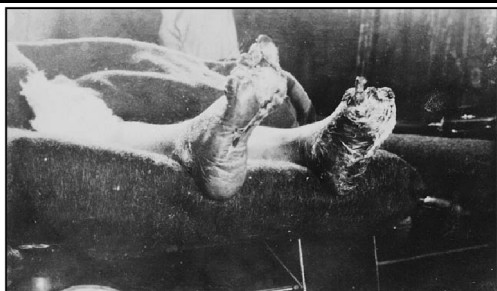
Detail:

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY ASSESSMENT

SOURCE A: CLINICAL MEDICAL RECORD PHOTOGRAPH OF SEVERE GANGRENE IN LOWER LIMB (C.1916)

SOURCE-A-TRENCH-FOOT-CLINICAL



RAMC clinical photograph documenting severe necrotic tissue damage and secondary gangrenous infection in the foot of a British soldier admitted to a base hospital from the frontline trenches.

Provenance: Official clinical photograph produced for the British Medical Research Committee / RAMC Case Records, c.1916. Wellcome Collection, London (Shelfmark: RAMC 442/1).

Historical Context: An official clinical photograph compiled for the Royal Army Medical Corps teaching archives c.1916, documenting the devastating progression of anaerobic necrosis and moist gangrene in a soldier's lower limb following prolonged exposure to contaminated Flanders mud and untreated tissue trauma.

Hinge Question: Study Source A. How does this clinical photograph illustrate the severity of anaerobic bacterial infections like gangrene when soil bacteria entered damaged, devitalized human flesh?

Study Source B. How useful is Source B for an enquiry into the methods used to protect British soldiers from poison gas attacks on the Western Front? Explain your answer, using Source B and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic British War Office Phenate-Hexamine (PH) anti-gas helmet manufactured in 1915–1916. What physical protective properties does chemically impregnated flannel possess, and what severe combat limitations did it have in battle?

STRUCTURE STRIP • 3-STEP UTILITY ANALYSIS

1. CONTENT & ARTIFACT UTILITY

Explain how the PH hood protected against chlorine and phosgene, featuring glass eyepieces and an exhalation valve.

2. PROVENANCE & MOTIVE

EVALUATION
Evaluate the artifact: official War Office standard issue, demonstrating rapid British industrial response to chemical warfare.

3. CONTEXT & COMBAT LIMITS

Cross-reference with knowledge: hot, suffocating, prone to fogging, and completely ineffective against mustard gas introduced in 1917.

ANALYTICAL CONNECTIVES:

Source B is useful for investigating gas defence because... • Specifically, the physical helmet proves that... • In terms of provenance, as a standard-issue War Office artifact... • However, Source B is limited because it cannot show... • Consequently...

KEY VOCABULARY BANK:

high-explosive artillery • shrapnel balls • gas gangrene • Clostridium welchii • Brodie steel helmet • chlorine gas (1915) • phosgene • mustard gas (1917) • PH helmet • Small Box Respirator (1916)

Task: Using the structure strip above, write your analytical source utility evaluation below:

TIMELINE MISSION

Sketch the Brodie steel helmet and British PH gas hood on Milestone 2

← Pages 2-3

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. How many stretcher bearers were assigned to an infantry battalion of 1,000 men?

2. How many bearers were typically required to carry one wounded man in Passchendaele mud?

3. How far behind the frontline was a Regimental Aid Post (RAP) usually located?

4. Could major surgery be performed at a Regimental Aid Post?

5. Which medical unit operated Advanced and Main Dressing Stations (ADS/MDS)?

6. What station in the evacuation chain was the primary surgical center near rail lines?

7. What three triage categories were used at Casualty Clearing Stations?

8. What female voluntary organisation drove motor ambulances on the Western Front?

9. Why were canal barges preferred for transporting chest and brain wound patients?

10. Where were large British Base Hospitals located in France?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain why the triage classification system at Casualty Clearing Stations was necessary when dealing with mass-casualty offensives.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

1(a) Describe one feature of the triage system used at Casualty Clearing Stations on the Western Front. [2 marks]

Hint: Explain how casualties were sorted into Walking Wounded, Immediate Surgery, and Moribund to maximize survival.

Feature:

Detail:

1(b) Describe one feature of the work of the First Aid Nursing Yeomanry (FANY) on the Western Front. [2 marks]

Hint: Consider ambulance driving, mobile soup kitchens, and transporting wounded through artillery fire.

Feature:

Detail:

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY mins]

SOURCE A: RAMC STRETCHER BEARERS CARRYING A CASUALTY AT PASSCHENDAELE (1917)

SOURCE-A-STRETCHER-BEARERS-PASSCHENDAELE



Four British stretcher bearers struggling through deep, liquid mud and waterlogged shell craters to evacuate a wounded soldier on a wooden stretcher near Boesinghe during the Third Battle of Ypres.

Provenance: Official British Army photograph taken by Lieutenant Ernest Brooks on the Western Front, Flanders, 1917. Imperial War Museum Collection (Shelfmark: Q 2757).

Historical Context: Official photograph taken by British war photographer Ernest Brooks in 1917 during the Third Battle of Ypres (Passchendaele), capturing the immense physical difficulty experienced by RAMC stretcher bearers navigating devastated, liquid-mud battlefields under continuous fire.

Hinge Question: Study Source A. How does this photograph illustrate why transporting wounded soldiers from the frontline to dressing stations took many exhausting hours in the Ypres sector?

Study Source A. How could you follow up Source A to find out more about the difficulties stretcher bearers faced evacuating casualties from the Western Front?

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

SENTENCE STEMS:

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

KEY VOCABULARY BANK:

RAMC • FANY • Regimental Aid Post • Advanced Dressing Station • Casualty Clearing Station • Base Hospital • triage system • motor ambulance • hospital train • canal barge

Enquiry Rationale: Explain in full sentences why your chosen source type is historically reliable and superior to alternatives:

TIMELINE MISSION

Draw the 6 sequential stages of the Chain of Evacuation on Milestone 5

← Pages 2-3

Motor ambulances: Guaranteed to rattle every uninjured bone in your body while speeding you to the Casualty Clearing Station.

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. What was the mortality rate for compound femur fractures in 1914 before the Thomas Splint?	2. Who introduced the Thomas Splint to frontline medical officers in December 1915?
3. To what percentage did the Thomas Splint reduce compound femur mortality by 1916?	4. How did the Thomas Splint prevent fatal internal haemorrhage in fractured thighs?
5. What surgical term describes cutting away dead, infected tissue from a gunshot wound?	6. What antiseptic irrigation solution was used continuously through perforated rubber tubes?
7. Why did Carrel-Dakin solution need to be made fresh in laboratories every 6 hours?	8. Who discovered X-rays in Germany in 1895?
9. Name one limitation of mobile X-ray units used at Casualty Clearing Stations.	10. What was the role of Marie Curie in mobile radiology during the Great War?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain how wound debridement combined with Carrel-Dakin irrigation prevented fatal gas gangrene in high-explosive wounds.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

1(a) Describe one feature of the Thomas Splint used on the Western Front. [2 marks]
Hint: Explain the padded metal ring fitted against the groin and the windlass traction system on the foot.
Feature: Detail:

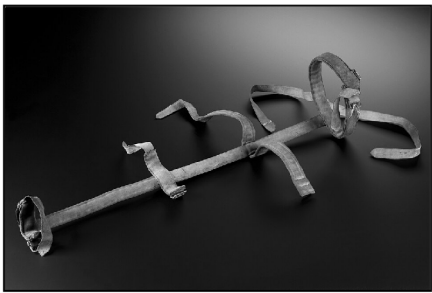
1(b) Describe one feature of the Carrel-Dakin method of treating infected wounds. [2 marks]
Hint: Explain the system of perforated rubber tubes irrigating the deep wound with sodium hypochlorite bleach.
Feature: Detail:

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY ASSESSMENT

SOURCE A: PHOTOGRAPH OF THE THOMAS SPLINT APPLIED TO A WOUNDED BRITISH SOLDIER (C.1916)

SOURCE-A-THOMAS-SPLINT-AUTHENTIC



A wounded British soldier in an RAMC field hospital with an authentic Thomas Splint applied to his fractured leg, showing the padded leather groin ring, rigid metal side bars, and extension traction cords.

Provenance: Official photograph produced for the RAMC Training and Photographic Archive, Western Front, c.1916. Wellcome Collection, London (Shelfmark: RAMC 512/3).

Historical Context: An official instructional photograph compiled for the Royal Army Medical Corps c.1916, illustrating the precise method of applying the Thomas Splint to a patient with a compound fracture of the femur. The padded ring fits firmly against the pelvis while traction is maintained on the foot, pulling the limb straight to prevent movement of broken bone fragments.

Hinge Question: Study Source A. How does the mechanical design of the Thomas Splint shown in the photograph explain why survival rates for fractured femurs rose from 20% in 1914 to 80% by 1917?

Study Source A. How useful is Source A for an enquiry into the methods used to treat wounded soldiers with fractured limbs on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic instructional photograph taken in an RAMC military hospital training ward c.1916. What does it demonstrate about mechanical traction and splinting, and why might an instructional training photograph differ from emergency frontline conditions in a mud dugout?

STRUCTURE STRIP • 3-STEP UTILITY ANALYSIS

1. CONTENT & UTILITY

Explain how Source A illustrates the Thomas Splint: groin ring, steel framework, and traction cord pulling the leg taut to prevent bone grinding.

2. PROVENANCE & TONE

Evaluate the photograph: produced for medical training manuals, demonstrating correct surgical application under clean, controlled conditions.

3. CONTEXT & LIMITS

Cross-reference with knowledge: Robert Jones 1915, 80% down to 20% mortality; note limitations—photograph cannot convey agony or muddy dugout application.

ANALYTICAL CONNECTIVES:

Source A is useful for an enquiry into limb treatments because... • Specifically, the photograph demonstrates... • In terms of provenance, as an RAMC instructional image... • However, the source is limited because... • Therefore...

KEY VOCABULARY BANK:

Thomas Splint • Hugh Owen Thomas • Robert Jones • compound fracture • femoral artery • mechanical traction • wound debridement • Carrel-Dakin solution • delayed primary closure • mobile X-ray unit

Task: Using the structure strip above, write your analytical source utility evaluation below:

TIMELINE MISSION

Sketch the Thomas Splint groin ring and traction cord on Milestone 3

← **Pages 2-3**

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. Who discovered the main ABO blood groups in Vienna in 1901?	2. What chemical was discovered by Albert Hustin in 1914 to prevent blood clotting?
3. What substance did Rous and Turner add in 1916 to preserve blood for up to 4 weeks?	4. Who established the world's first stored blood depot at the Battle of Cambrai in 1917?
5. What blood group was chosen for universal transfusion in Robertson's blood depot?	6. Which American surgeon pioneered precise neurosurgery using local anaesthetic on the Western Front?
7. To what percentage did Harvey Cushing reduce brain surgery mortality in 1917?	8. Which New Zealand-born surgeon pioneered facial reconstructive plastic surgery at Sidcup?
9. What surgical skin-transfer technique did Harold Gillies invent in 1917?	10. What hospital in Kent became the world centre for British plastic facial reconstruction?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain how sodium citrate combined with refrigeration revolutionized blood transfusion between 1914 and 1917.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

1(a) Describe one feature of the blood depot established by Oswald Robertson at the Battle of Cambrai in 1917. [2 marks]

Hint: Consider the use of sodium citrate, ice chests, Type O blood, and treating shock before surgery.

Feature:	Detail:
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1(b) Describe one feature of the surgical methods used by Harold Gillies at Queen's Hospital, Sidcup. [2 marks]

Hint: Explain the tubed pedicle method of maintaining blood supply while grafting facial tissue.

Feature:	Detail:
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Question 2(b): Follow-Up Enquiry Table [4 marks • 5 PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY mins]

SOURCE A: CAPTAIN OSWALD ROBERTSON'S BLOOD STORAGE AND TRANSFUSION APPARATUS AT CAMBRAI (1917)

SOURCE-A-ROBERTSON-BLOOD-DEPOT-1917



The authentic wooden ice chest, glass collection flasks, rubber delivery tubing, and sodium citrate solutions utilized by Captain Oswald Robertson to establish the world's first blood bank during the Battle of Cambrai in November 1917.

Provenance: Primary medical artifact preserved in the Royal Army Medical Corps Museum / Wellcome Historical Medical Collection (Accession: RAMC 628/1).

Historical Context: An authentic surviving artifact photograph of Captain Oswald Robertson's blood storage and transfusion kit used at the Battle of Cambrai in November 1917. Robertson collected blood in sterile glass bottles containing sodium citrate and dextrose (glucose) and stored them inside an insulated wooden ice chest, enabling stored blood to be transfused up to 26 days after collection.

Hinge Question: Study Source A. How does this blood storage apparatus explain how medical officers at the Battle of Cambrai were able to treat soldiers suffering from severe shock before operations began?

Study Source A. How could you follow up Source A to find out more about the success of stored blood transfusions at the Battle of Cambrai in 1917?

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

SENTENCE STEMS:

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

KEY VOCABULARY BANK:

Karl Landsteiner • ABO blood groups • sodium citrate (Hustin 1914) • glucose preservative (Rous & Turner 1916) • Oswald Robertson • Battle of Cambrai (1917) • blood depot • Harvey Cushing • Harold Gillies • tubed pedicle graft

Enquiry Rationale: Explain in full sentences why your chosen source type is historically reliable and superior to alternatives:

TIMELINE MISSION

Sketch Oswald Robertson's iced blood chest and Harold Gillies' tubed pedicle on Milestone 6

← Pages 2–3

Harold Gillies at Sidcup: Turning shattered faces into men again with tubed pedicle skin grafts, while patients politely pretended not to notice.

STUDENT ASSESSMENT RECORD & PROGRESS TRACKER

PUPIL NAME: _____ CLASS: _____ TARGET GRADE: _____

Date Completed	Lesson • Specification Focus	Q1 Feature Drills [4m]	Q2 Source Practice	Lesson Total
	KT5.1: Trench System & Battles	Q1: [/ 4]	Q2a Util: [/ 8]	[/ 12]
	KT5.2: Trench Illnesses & Mud	Q1: [/ 4]	Q2b Follow: [/ 4]	[/ 8]
	KT5.3: Combat Trauma & Gas	Q1: [/ 4]	Q2a Util: [/ 8]	[/ 12]
	KT5.4: Evacuation Chain & CCS	Q1: [/ 4]	Q2b Follow: [/ 4]	[/ 8]
	KT5.5: Splints, Debridement, X-Rays	Q1: [/ 4]	Q2a Util: [/ 8]	[/ 12]
	KT5.6: Blood Storage & Reconstruction	Q1: [/ 4]	Q2b Follow: [/ 4]	[/ 8]
SECTION A ASSESSMENT TOTALS		Q1 Total: [/ 24]	Q2 Total: [/ 36]	[/ 60]

TEACHER FORMATIVE ASSESSMENT • FEEDBACK

Effort: [1 • 2 • 3 • 4 • 5]

WHAT WENT WELL (WWW):

EVEN BETTER IF (EBI):

Teacher Signature: _____Date: _____

INTERACTIVE DIGITAL QUIZZING HUB • SCAN FOR INSTANT RETRIEVAL PRACTICE

ONLINE RECALL

KT5.1

Trenches & Mud

SCAN TO QUIZ

Score: [/ 8]

KT5.2

Trench Illness

SCAN TO QUIZ

Score: [/ 8]

KT5.3

Trauma & Gas

SCAN TO QUIZ

Score: [/ 8]

KT5.4

Evacuation

SCAN TO QUIZ

Score: [/ 8]

KT5.5

Splints & X-Rays

SCAN TO QUIZ

Score: [/ 8]

KT5.6

Blood & Plastic

SCAN TO QUIZ

Score: [/ 8]